FEUP

FACULTY OF ENGINEERING - UNIVERSITY OF PORTO

Systems Engineering

Requirements Specification and Architecture

SWAID - Global Document

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1 Introduction

The Standing Waves Acoustic Interactive Demonstrator (SWAID) is an integrated installation merging physics, music, and visual arts. It visualizes acoustic standing waves by exciting a Chladni plate to form complex geometric powder patterns. These patterns, along with synchronized lighting, are modulated in real-time via user hand gestures and musical parameters.

1.1 Document Presentation

This document serves as the Global Requirements Specification and Architecture baseline for SWAID. It integrates the constraints from the three development sub-teams (CP, PR, and HI) to ensure strict alignment, establish clear system boundaries, and define critical inter-team dependencies.

1.2 Terms and Definitions

SWAID Standing Waves Acoustic Interactive Demonstrator.

CP Chladni Plate Subsystem (enclosure, transducers, power, and lighting).

PR Plate Resonance Subsystem (signal generation, temporal sync, and pattern mapping).

HI Human Interface Subsystem (computer vision, audio synthesis, and GUI).

Chladni Plate Tuned metal sheet used to visualize standing waves via accumulated powder.

Transducer Electromechanical actuator used to excite the plate's resonance frequencies.

Resonance Alphabet PR library mapping visual targets to specific transducer parameters.

Settling Time Physical time required for the powder to stabilize into a new pattern.

API Standardized software communication bridge between the HI and PR subsystems.

GUI Graphical User Interface generated by HI and projected onto the CP hardware.

1.3 Changes

Version	Date	Author	Description
1	27th Feb	Lucas Aparício	Template creation
2	20th Mar	M. Amorim, R. Oliveira, D. Silva	First Version

2 System General Perspective

2.1 Inter-Team Dependencies:

- **Enclosure Sizing and Mounting (CP ← PR):** The physical properties of the Chladni plate (dimensions, material, and thickness) have been explicitly defined by the PR subsystem to optimize the required acoustic resonance modes. Consequently, the CP team's mechanical design is strictly dependent on these specifications; the CP subsystem must design the structural enclosure, transducer mounting chassis, and powder containment box to precisely accommodate and support the PR-selected plate without restricting its vibrational freedom.
- **Settling Time vs. Command Rate (PR ← HI):** While the PR electronic system can switch signals in milliseconds, the physical inertia of the powder requires a settling time to form stable images. *Mitigation:* The HI team must implement a "state-hold" or "debounce" logic in their software to prevent switching symbols faster than the physical medium can respond.
- **Gesture-to-Symbol Latency (PR ← HI):** The "Real-Time" attractiveness of the demonstrator relies on the HI team's image processing speed. The PR team's low-latency performance is only effective if the HI team provides modulation commands with minimal preprocessing delay.
- **Interface Contract (PR ⇔ HI):** The PR and HI teams must mutually define the software development stack and the data exchange format early in the development cycle. The PR team is dependent on the HI team adhering to the established API definitions to ensure successful integration of the "Resonance Alphabet" commands. Unilateral changes to this interface by either team will compromise the subsystem's integration.
- **Settling Time vs. Command Rate (HI ↔ PR):** While the PR electronic signal generator can switch frequencies in milliseconds, the physical inertia of the Chladni powder requires a settling time to form stable images. The PR team depends on the HI team to implement a "state-hold" or "debounce" logic in their system, ensuring that user gestures do not trigger pattern changes faster than the physical medium can respond.
- **Visual Output and Display (HI → CP):** The HI team's graphical user interface (GUI) and volume visualizations are purely software-based. The HI team depends on the CP team to provide, wire, and power the physical LED matrix displays and LED strips required to project these interfaces to the user.
- **Acoustic Isolation (HI ← CP):** The primary auditory experience is dictated by the HI team's synthesized musical audio playing through system speakers. The HI team depends on the CP team to properly soundproof the structural enclosure to attenuate the harsh, high-frequency mechanical noise generated by the transducers, preventing it from overpowering the musical output.

3 Unified SWAID Project Requirements Specification

This table consolidates the refined and aligned requirements for the three integrated subsystems: Chladni Plate (CP), Plate Resonance (PR), and Human Interface (HI).

Table 1: Consolidated Requirements Specification (CP, PR, and HI)

ID	Type	Priority	Description
Chladni Plate (CP) Subsystem			
CP-01	Functional	High	The CP subsystem shall provide programmable RGB LED lighting to enhance user interactive and visual experience.
CP-02	Internal Interface	High	The CP lighting hardware shall provide a direct digital control interface to receive synchronized illumination commands from the central processing units.
CP-03	Design Constraint	Medium	The CP structural enclosure shall incorporate a mechanical access mechanism (e.g., a drawer) to facilitate electronics maintenance and rapid Chladni powder refilling without full disassembly.
CP-04	Design Constraint	High	The CP subsystem shall integrate exactly four acoustic transducers physically coupled to the plate for excitation.
CP-05	Design Constraint	Medium	The total mass of the fully assembled SWAID system shall not exceed 10.0 kg.
CP-06	Performance	High	The CP acoustic transducers shall sustain continuous operation at maximum rated resonance frequencies for a minimum of 8 hours without thermal degradation or performance loss.
CP-07	Design Constraint	High	The total cumulative material cost for the entire SWAID system (all three subsystems) shall not exceed 480 EUR.
CP-08	Design Constraint	Medium	The Chladni plate shall be a 200 mm $\times$ 300 mm, 1 mm thick stainless steel sheet with an 8k mirror polished finish.
CP-09	External Interface	Medium	The CP subsystem shall interface with external facility power via a single standard AC power cable to supply the entire system.
CP-10	External Interface	Medium	The system shall provide a physical master ON/OFF switch on the external enclosure to allow the user to control the main power supply.
CP-11	Functional	Medium	The CP subsystem enclosure shall incorporate a physical containment barrier or reservoir to capture and store powder that falls off the plate edges.
CP-12	Functional	Medium	The CP subsystem shall integrate a programmable LED matrix display on the rear interior wall to visualize the graphical user interface provided by the HI subsystem.

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ID	Type	Priority	Description
CP-13	Functional	Medium	The CP subsystem shall include two vertically oriented LED strips adjacent to the rear LED display to serve as dynamic audio volume indicators.
CP-14	Design Constraint	Medium	The CP enclosure shall incorporate a one-way mirror positioned above the primary steel plate to generate a visual infinity mirror effect.
CP-15	Design Constraint	Medium	The CP subsystem shall utilize UV-fluorescent powder and integrate UV/Blacklight LEDs to produce a glowing effect within the infinity mirror setup.
CP-16	Design Constraint	High	The CP internal power distribution network shall convert the primary AC input into stable, isolated DC voltage rails as required by transducers, LEDs, speakers, and computational hardware.
CP-17	Performance	High	The CP structural enclosure shall provide acoustic isolation to significantly attenuate transducer operational noise, ensuring it does not interfere with the external speaker audio.
CP-18	Design Constraint	Medium	The CP enclosure shall feature an opaque top cover to prevent ambient environmental light from degrading the internal UV and LED illumination effects.
CP-19	Design Constraint	Medium	The CP enclosure architecture shall consist of two separable modular units: a primary base containing the electronics and plate, and a removable top transparent acrylic case with an opaque ceiling and rear wall.
Plate Resonance (PR) Subsystem			
PR-01	Functional	High	The PR subsystem shall generate independent, synchronized actuation signals, specifying frequency, phase, and amplitude, for the acoustic transducers.
PR-02	Functional	High	The PR subsystem shall maintain an internal mapping library to directly translate high-level target pattern commands into precise transducer actuation parameters.
PR-03	Functional	Medium	The PR subsystem shall generate illumination control signals that are temporally synchronized with the active Chladni resonance mode state.
PR-04	Performance	High	The PR subsystem processing latency, measured from the receipt of an HI command to the output of transducer actuation signals, shall not exceed 50 milliseconds.
PR-05	Performance	Medium	The PR subsystem shall ensure strict temporal synchronization between acoustic actuation signals and lighting control commands, with jitter not exceeding 10 milliseconds.

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ID	Type	Priority	Description
PR-06	Quality Attribute	Medium	The PR signal generation architecture shall be abstracted (alphabet-agnostic) to support dynamic re-mapping and expansion of the resonance library without modifying core generation logic.
PR-07	Functional	High	The PR subsystem firmware shall enforce digital limits on maximum signal amplitude and duty cycle to prevent thermal overloading and damage to the CP transducers.
PR-08	Design Constraint	Medium	The PR subsystem processing hardware shall conform to the dimensional constraints of the electronics compartment designated by the CP subsystem.
PR-09	Internal Interface	High	The PR subsystem shall expose a software API to accept standard high-level command strings (e.g., notes, chords) from the HI subsystem.
PR-10	Internal Interface	High	The PR subsystem shall transmit low-level actuation parameters to the CP transducer drivers via a standardized digital communication protocol.
PR-11	Performance	Low	The PR subsystem shall initialize, load the resonance dictionary into memory, and achieve a ready state within 10 seconds of receiving power.
PR-12	Functional	High	The PR subsystem resonance mode library shall contain a sufficient number of distinct, stable Chladni patterns to support a comprehensive mapping of HI musical notes and chord configurations.
Human Interface (HI) Subsystem			
HI-01	Functional	High	The HI subsystem shall capture overhead video of the user and apply computer vision algorithms to detect and track manual hand gestures in real-time.
HI-02	Functional	Medium	The HI subsystem shall translate recognized user hand gestures into specific musical commands, functioning as an interactive musical instrument.
HI-03	Performance	Medium	The system-wide response latency, from the occurrence of a physical gesture to the resulting visual pattern change on the Chladni plate, shall not exceed 1.0 second.
HI-04	Design Constraint	High	The HI subsystem shall utilize a dedicated, locally integrated processing unit to execute all computer vision and audio synthesis computations.
HI-05	Quality Attribute	High	The HI subsystem software setup and physical calibration routine shall require less than 5 minutes for a single operator to initialize.

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ID	Type	Priority	Description
HI-06	Quality Attribute	High	The HI subsystem interactive gesture vocabulary shall be intuitive, requiring no prior training or technical documentation for general public use.
HI-07	Functional	High	The HI subsystem shall feature an autonomous demonstration mode that sequentially plays pre-recorded audio and visual patterns when no user interaction is detected.
HI-08	Functional	Medium	The HI subsystem shall generate a graphical user interface (GUI) to provide visual feedback of the instrument state, outputting the video signal to the CP rear display.
HI-09	Functional	High	The HI subsystem software shall map every distinct musical note or chord to a corresponding Chladni symbol request sent to the PR subsystem.
HI-10	Functional	Medium	The HI subsystem shall allow the user to dynamically modulate audio parameters, specifically master volume, base frequency, and tempo, via distinct gestures.
HI-11	Internal Interface	Medium	The HI subsystem shall continuously transmit real-time volume state data to the CP lighting controller to drive the vertical LED volume indicators.
HI-12	Functional	High	The HI subsystem shall synthesize the resulting musical audio signal and output it directly to the integrated system speakers.

4 System Architecture

4.1 [CP] Hardware Design

TODO:

4.2 [CP] 3D Model Design

TODO:

4.3 [PR] Software Architecture

TODO:

4.4 [HI] Software Architecture

TODO: